CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 12: Surface Areas and Volumes

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two cubes, each of volume 125 cm^3, are joined end to end. Find the surface area of the resulting cuboid.

2. A vessel is a hollow cylinder with a hollow hemisphere of the same radius 7 cm joined below it. The total height of the vessel is 17 cm. Find its inner surface area.

3. A toy is a cone of radius 3 cm mounted on a hemisphere of the same radius. The total height of the toy is 7 cm. Find its total surface area in terms of π.

4. A cubical block of side 14 cm has a hemisphere placed on its top face. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.

5. A solid is a cone of height 4 cm standing on a hemisphere, both of radius 3 cm. Find the volume of the solid in terms of π.

6. A cylinder of height 10 cm and radius 6 cm has a conical cavity of height 8 cm and the same radius hollowed out. Find the volume of the remaining solid in terms of π.

7. A tent is a cylinder of height 3 m and radius 7 m with a conical top of slant height 10 m. Find the area of canvas used.

8. A solid pole is a cylinder of height 200 cm and base radius 10 cm, with a second cylinder of height 50 cm and radius 6 cm fixed on top. Find the volume of the pole. Take $\pi = 3.14$.

9. A sphere of radius 3 cm is dropped into a cylindrical vessel of radius 6 cm that is partly filled with water. If the sphere is fully immersed, by how much does the water level rise?

10. A wooden article is made by scooping out a hemisphere from each end of a solid cylinder of height 12 cm and base radius 2.8 cm. Find the total surface area of the article.

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Answer Key

1. Side 5 cm; cuboid $10 \times 5 \times 5$; surface area $= 250$ cm²
2. Cylinder height 10 cm; $2\pi r(h + r) = 2 \times \frac{22}{7} \times 7 \times 17 = 748$ cm²
3. Cone height 4 cm, $l = 5$ cm; $\pi r l + 2\pi r^2 = 15\pi + 18\pi = 33\pi$ cm²
4. 14 cm; $6 \times 196 + \frac{22}{7} \times 49 = 1330$ cm²
5. $12\pi + 18\pi = 30\pi$ cm³
6. $360\pi - 96\pi = 264\pi$ cm³
7. $\pi r(2h + l) = \frac{22}{7} \times 7 \times 16 = 352$ m²
8. $3.14 \times (20000 + 1800) = 68452$ cm³
9. $\frac{36\pi}{36\pi} = 1$ cm
10. $2\pi r(h + 2r) = 2 \times \frac{22}{7} \times 2.8 \times 17.6 = 309.76$ cm²

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